

THORNVEIL LLC

Sovereign AI infrastructure for federal missions.

IL5-READY • ECCN-AWARE • GOVERNANCE-FIRST

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EVALUATION ENGAGEMENTS ACTIVE

Ten engineering systems for sovereign federal AI — distributed inference, classification-gated chat, autonomous red team, capability gates, perception, and memory-safe firmware. Built ground-up to clear FedRAMP, IL5, and ECCN review on day one. Shipped solo while active-duty USMC.

CORE COMPETENCIES

Mycelium — distributed substitute-on-failure inference; keeps generating when half the mesh dies. 250K Go, 50-endpoint OpenAPI, mTLS, HMAC-chained audit. v1.0.0 cosign-signed.

RigRun — classification-gated AI chat; CUI+ hard-blocked from cloud at the type-system level. Three apps, one Go backend. 44 NIST 800-53 controls as code paths.

Auspex — autonomous AI red team gated by Signet; scope-as-code, 13 authority checks, HMAC + RFC 3161 audit. EAR ECCN 4D004.

Signet — capability-based safety gates for LLM agents; 4-stage / 11-check pipeline, RFC 3161 anchoring. Apache-2.0, on PyPI.

HawkStack — sub-million-parameter perception; 15 verified checkpoints across 5 domains, power-law scaling fit $R^2=0.9895$.

Meridian — memory-safe Rust flight-control firmware; 98K LOC / 60 crates, ed25519 airworthiness attestation, GPS-spoof / RC-hijack / mag-jam failsafes.

Canopy — multi-modal counter-UAS; RF + acoustic + thermal + Remote ID + ADS-B fusion, Kalman + Dempster-Shafer + bearing geolocation.

Alchemist — algorithm-aware C-to-Rust translation; five mandatory correctness gates, local-LLM only. Apache-2.0.

Pyros — self-tuning inference engine; 24 adaptive subsystems (PID + Holt + UCB1 + ϵ -greedy), race-clean.

Navigator — internal AI agent factory; 14-step pipeline, 9 production agents, 80 MCP-exposed tools.

DIFFERENTIATORS

- **Federal-fit by design** — built to clear FedRAMP, IL5, and ECCN review on day one, not retrofitted after the ask.
- **Cryptographically auditable** — every load-bearing system ships an HMAC-chained, RFC-3161-anchored, replayable audit a 3PAO can verify.
- **Sovereign / air-gap capable** — runs on customer hardware; no data leaves the building, no per-token cloud dependency.
- **Open-source verifiable** — public companion repos for evaluators; two Apache-2.0 releases on GitHub and PyPI.

PAST PERFORMANCE

- **First autonomous AD takeover by an AI red team** — Domain Admin under strict autonomy in 4m29s; HMAC-chained audit, signed-verifier VALID (2026-05-16).
- **Mycelium substitute-on-failure RTP** — reduction-to-practice with tamper-evident audit log; cosign-signed v1.0.0 (2026-05-08).
- **Signet** — Apache-2.0 production release on PyPI as signet-sign; 11 production checks, RFC 3161 anchoring.

FOUNDER

Jesse Morgan — USMC Staff Sergeant, Signals Intelligence, TS/SCI. Eight years active duty, six countries deployed, national-level intelligence reporting under classified handling. BA Arabic, Defense Language Institute. Built all ten systems solo, January–May 2026.

COMPANY DATA

Entity	Thornveil LLC	UEI	NCBZNKMJB59
Location	Tampa, Florida	CAGE	1T0K4
SAM.gov	Active • All Awards	NAICS	541511, 541512, 334511
Business type	Small business • VOSB-eligible 12/2026	Vehicles	Direct • SBIR I/II • OTA • CRADA
Contracts	Open 2026-12-10 • FAR 12.212 commercial item	Engage	Eval NDA • replay pkg • walkthrough

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